MOHAN BABU UNIVERSITY

DATA SCIENCE LAB — SAMPLE EXPERIMENT

Experiment: Machine Learning

Student Name: _____

Student ID: _____

Date: _____

Aim

To train and evaluate a basic supervised machine learning model.

Requirements

Python, Pandas, and scikit-learn.

Experiment

1. Load a small labeled dataset, such as Iris, using scikit-learn.
2. Inspect feature columns and target labels.
3. Split the data into training and testing sets using train_test_split.
4. Train a Decision Tree Classifier.
5. Predict labels for the test set.
6. Calculate accuracy and display a confusion matrix.
7. Write a brief note explaining why the model is evaluated on test data.

Expected Output

Trained model predictions, accuracy score, and confusion matrix

Conclusion

Summarize the model's performance and one limitation

Note: This is a sample practice lab experiment created for learning and website testing. It is not an official university-issued lab manual.